

QIntern 2026 Project Proposal

Title

Quantum Correlations and Bell Nonlocality in Continuous-Variable Gaussian Systems

Project Category

Quantum Information / Quantum Optics / Continuous-Variable Quantum Computing

Mentor

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Project Description

Quantum correlations constitute essential resources for modern quantum technologies, including quantum communication, quantum cryptography, quantum sensing, and quantum computation. In continuous-variable (CV) quantum systems, Gaussian states generated through optical processes provide experimentally accessible platforms for investigating nonclassical phenomena such as entanglement, Bell nonlocality, quantum steering, and squeezing.

This project aims to investigate quantum correlations in two-mode Gaussian systems inspired by nondegenerate three-level lasers and nonlinear optical platforms. The participants will study the generation and characterization of quantum correlations through covariance matrix techniques and phase-space methods.

Particular attention will be devoted to Bell inequality violation in continuous-variable systems and its relation to other quantum correlation measures such as entanglement and Gaussian steering. The project will also explore how dissipation, squeezing, and environmental effects influence the nonclassical properties of the generated states.

The internship combines theoretical quantum optics, quantum information theory, and numerical simulations using Python and quantum computing libraries.

Objectives

The main objectives of the project are:

1. Introduce participants to continuous-variable quantum information.
2. Study Gaussian states and covariance matrix formalism.
3. Simulate two-mode quantum optical systems using Python.
4. Analyze entanglement and Bell inequality violation.
5. Explore the effect of squeezing and dissipation on quantum correlations.
6. Develop numerical and visualization tools for quantum-state analysis.
7. Prepare a short scientific report and presentation.

Expected Outcomes

By the end of the internship, participants are expected to:

- Understand the fundamentals of Gaussian quantum states.
- Learn how to compute covariance matrices and symplectic eigenvalues.
- Implement numerical simulations for continuous-variable systems.
- Evaluate Bell inequality violation and entanglement measures.
- Gain experience with scientific programming tools used in quantum information.
- Produce reproducible numerical results and visualizations.

Potential outcomes may include:

- Open-source simulation notebooks.
- A technical report or preprint.
- Future conference or publication opportunities.

Required Background

Preferred background includes:

- Undergraduate-level quantum mechanics.
- Linear algebra and basic probability.
- Basic programming knowledge in Python.

Familiarity with quantum information or quantum optics is beneficial but not mandatory.

Tools and Technologies

Participants will work with:

- Python
- NumPy
- SciPy
- Matplotlib
- QuTiP
- Jupyter Notebook

Optional:

- Qiskit
- Strawberry Fields

Tentative Work Plan

Week 1

- Introduction to quantum information and Gaussian states
- Python setup and numerical tools
- Review of covariance matrix formalism

Week 2

- Simulation of two-mode Gaussian systems
- Calculation of quantum correlations
- Visualization of covariance matrices and phase-space quantities

Week 3

- Bell inequality violation in CV systems
- Effect of squeezing and dissipation
- Numerical parameter exploration

Week 4

- Final simulations and optimization
- Preparation of scientific report and presentation
- Discussion of future research directions

Scientific Motivation

Continuous-variable quantum systems provide one of the most experimentally relevant frameworks for quantum technologies due to their compatibility with optical platforms such as optical parametric oscillators and squeezed-light sources. Understanding the interplay between Bell nonlocality, entanglement, and environmental effects is crucial for developing robust quantum-information protocols and for advancing future photonic quantum technologies.

This project offers participants an opportunity to explore modern topics at the interface of quantum optics, quantum information, and numerical simulation.

References

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2. H. M. Wiseman, S. J. Jones, and A. C. Doherty, “Steering, entanglement, nonlocality, and the Einstein–Podolsky–Rosen paradox,” *Phys. Rev. Lett.* **98**, 140402 (2007).
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4. K. Talbi *et al.*, “Witnessing Entanglement via Extractable Work in a Nondegenerate Three-Level Laser,” *Open Systems & Information Dynamics* **32**, 2550003 (2025).
5. J. El Qars, “Gaussian Rényi-2 correlations in a nondegenerate three-level laser,” *Quantum Information Processing* **23**, 83 (2024).